Ex. No.: 1 **Date:** 26/12/24

Calculate Area and Perimeter

Write an Algorithm and draw a Flowchart to Calculate the area and perimeter of a square.

Algorithm:

Step 1: Input the side of the square.

Step 2: Multiply the side by side for area.

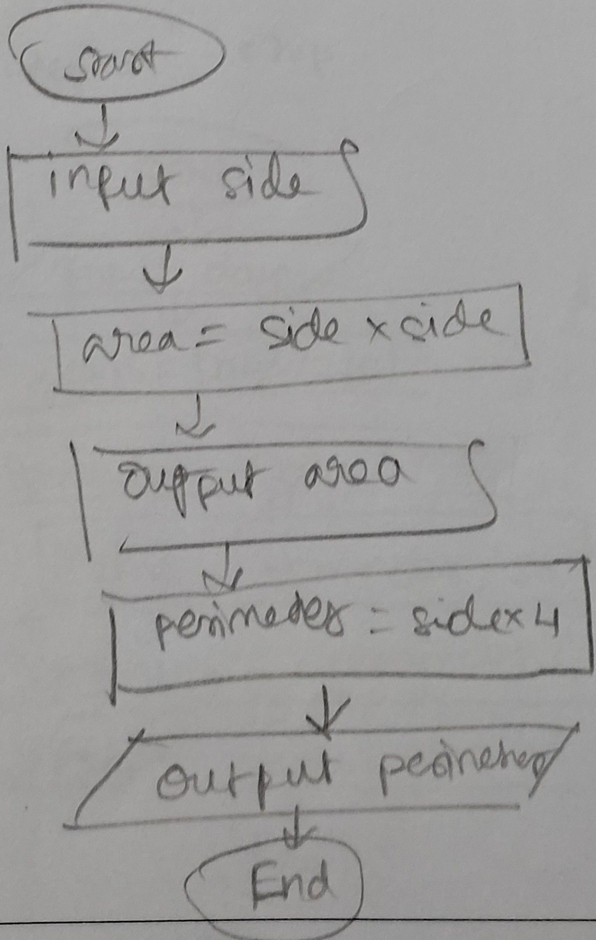
Step 3: print the area

Step 4: Multiply the side by 4 perimeter

Step 5: print the perimeter.

Step 6: End

Flowchart:



Ex. No.: 2 **Date:** 25/12/24

Days to Year Conversion

Write an Algorithm and draw a Flowchart to convert the given days into years & months.

Algorithm:

step 1 : start

step 2 : Input the no. of years, read days

step 3 : calculate the no. of years

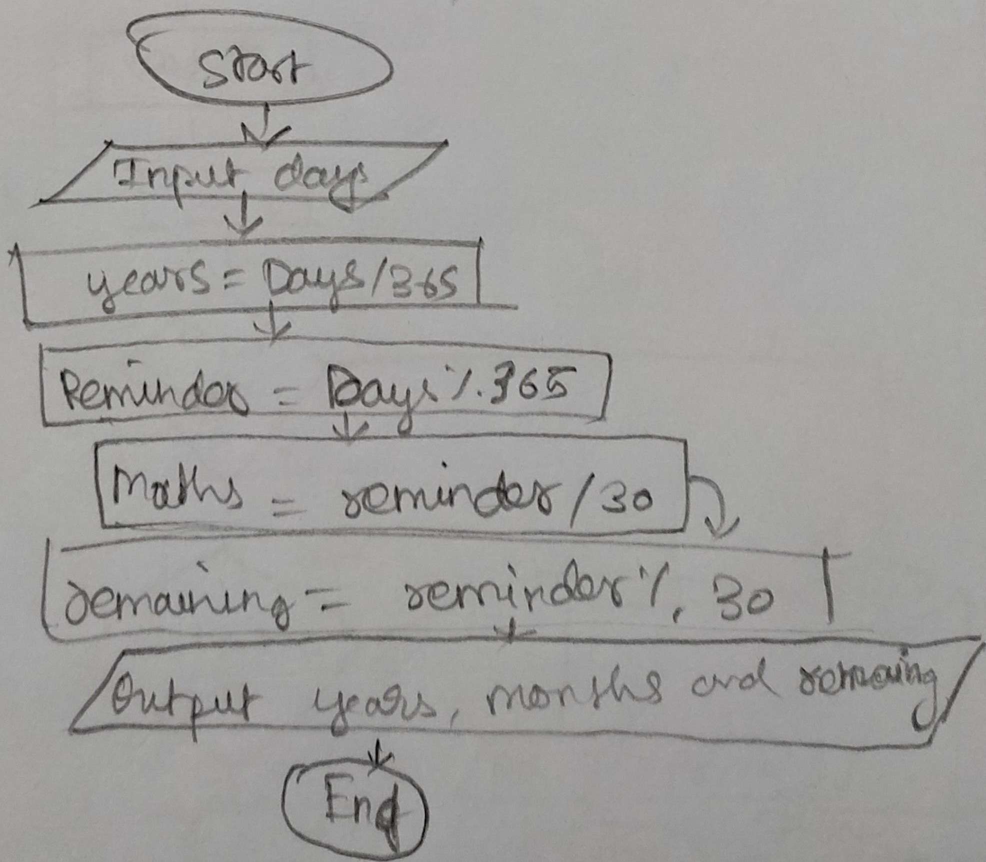
years = days // 365.

step 4 : Calculate the remaning days after years & remaining days = days %30.

step 5 : Print years and months

Flowchart:

step 6 : stop



Ex. No.: 3

Date: 28/12/24

Prime Number

Write an Algorithm and draw a Flowchart to check whether the given number is Prime or not.

Algorithm:

1: Start

2: Read int value num

3: while i<num

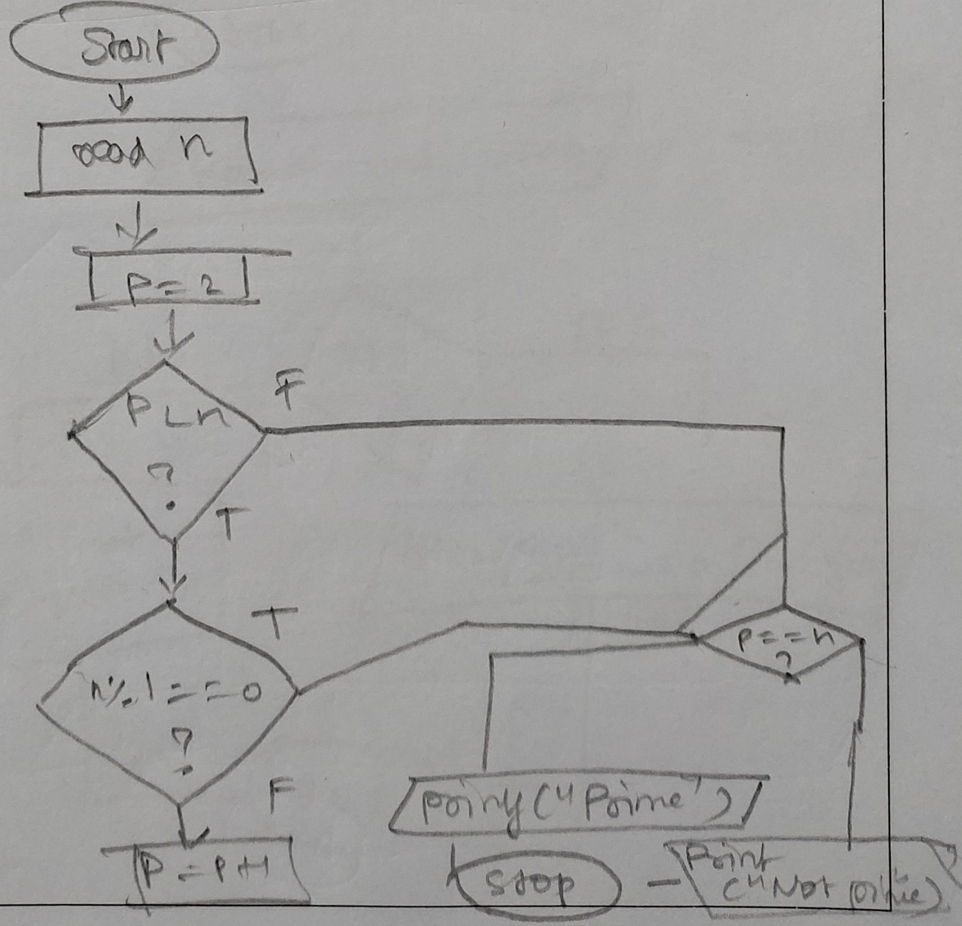
4: if (num % i = 0)

5: print "It is not prime no".

6: update i value

7: else print ("It is prime no")

Flowchart:



Ex. No.: 4 **Date:** 26/12/24

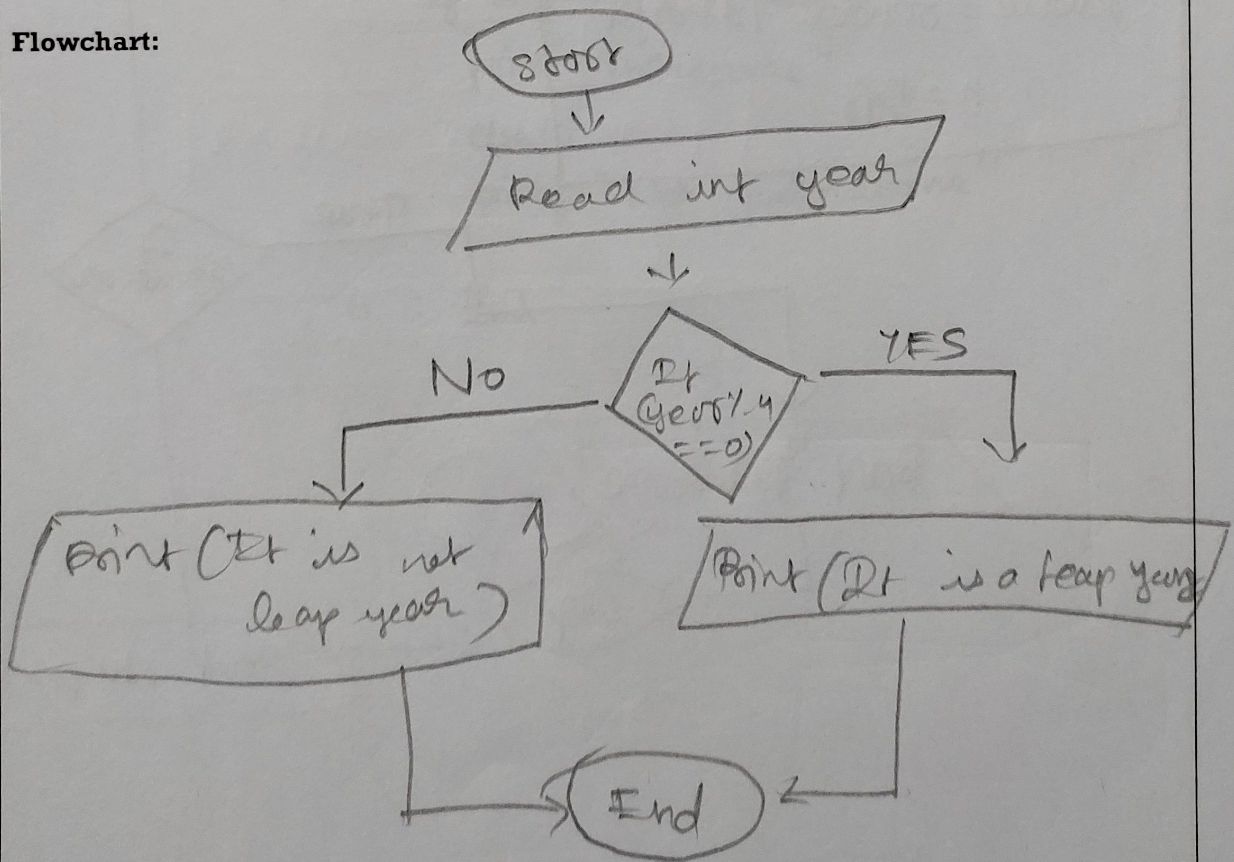
Leap Year

Write an Algorithm and draw a Flowchart to check whether the given year is Leap year or not.

Algorithm:

1. start
2. Read int year
3. if (a%4 ==0)
4. Print It is a leap year.
5. else print "it is not a leap
6. yare"
6. stop

Flowchart:



Ex. No.: 5 **Date:** 28/12/24

Palindrome Number

Write an Algorithm and draw a Flowchart to check whether the given number is palindrome number or not.

Algorithm: Step 1: start

2: Read the number n

3: Initialyze set original = n reversed = 0

4: while n>0

- set digit = n mod 10

update reversed = reversed x 10 + digit

- update n = n ÷ 10

5: if original = reversed

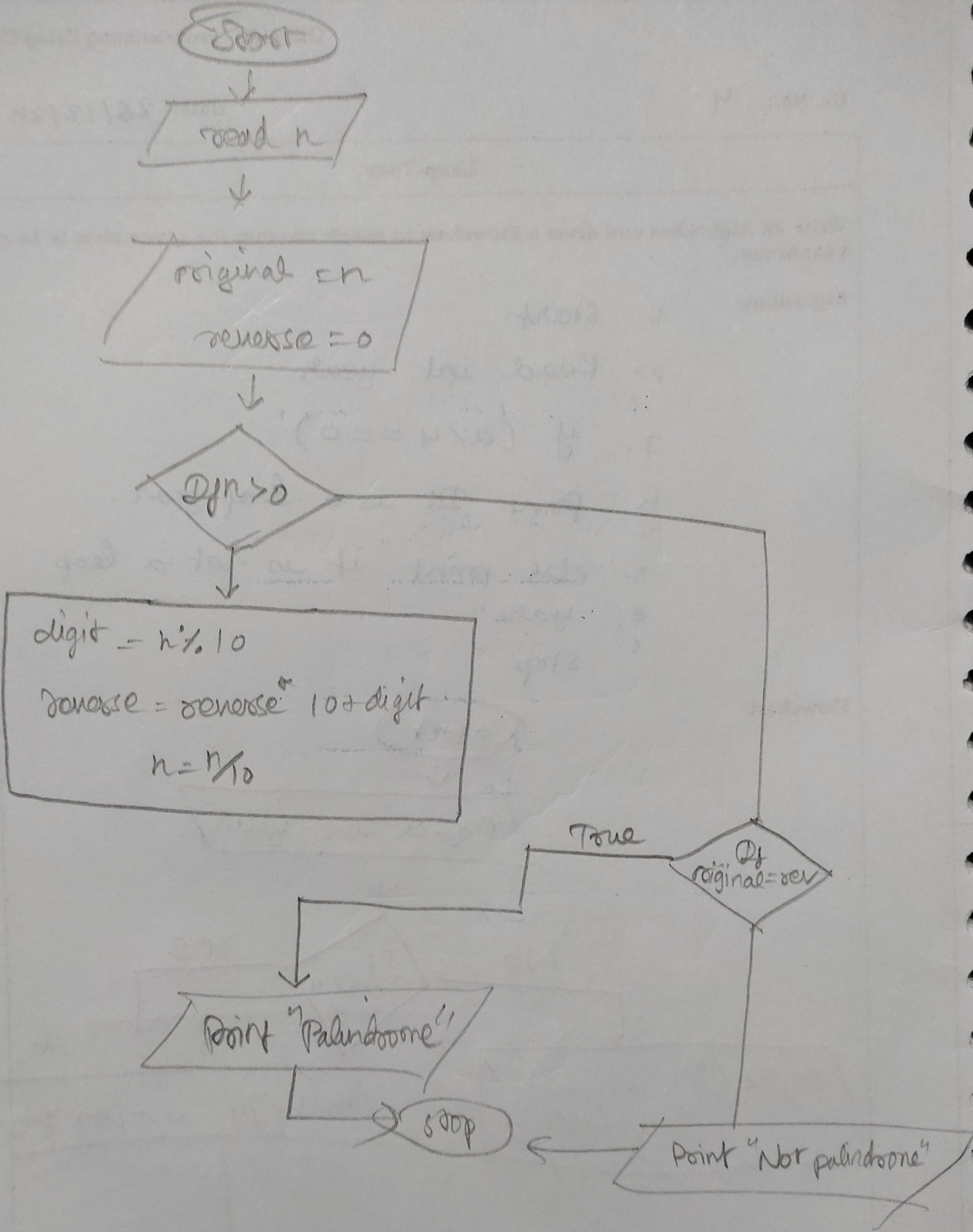
print "Palindrome"

else

printf "Not palindrome"

6: End

Flowchart:


Start
read n
original = n
reverse = 0
If n > 0
digit = n % 10
reverse = reverse * 10 + digit
n = n/10
True
If original = rev
Print "Palindrome"
Stop
Print "Not palindrome"

Ex. No.: 6 **Date:** 26/12/24

Sum of Digits

Write an Algorithm and draw a Flowchart to calculate the sum of digits in the given number.

Algorithm: Step -1: start

2: Input the number (n)

3: Initialyze s=0

4: Repeat the following steps while n is greater than 0. n>0 - Extract the last digit

digit =n% 10

- Add the digits is sum

sum= sum+digit

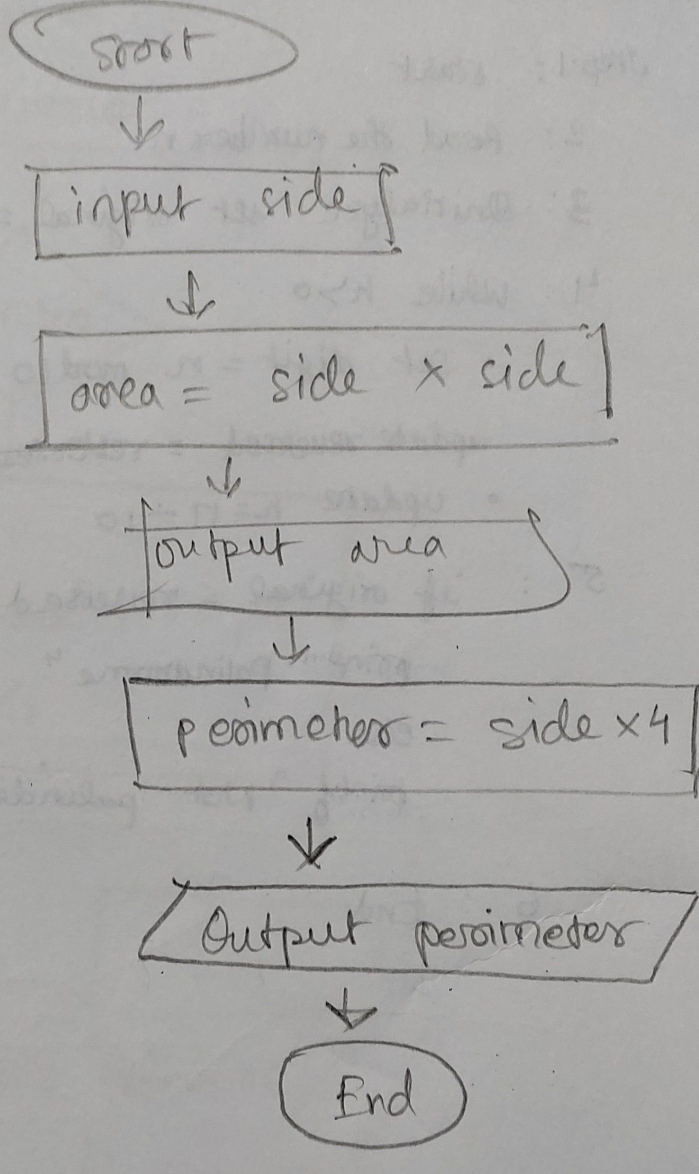
- Remove the last digit from n;

n← n//10

5: Output the sum.

6: End.

Flowchart:


start
input side
area = side x side
output area
perimeter = side x 4
Output perimeter
End